



InsightForge AI — Data Analysis Report

Generated by InsightForge AI · June 26, 2026



Key Performance Indicators

10,108	26	0.0%	73901042
Total Rows	Total Columns	Missing Data %	3.58
			Mean Client_Num

Data Preview

Client Num	Ann Dates	Activation Dates	Customer Acquisition Dates	Week Start Dates	Week End Dates	Credit Limits	Total Reviews	Total Ratings	Total Transactions	Avg Utilization Ratio	Interests Earned	Delinquency Rate	Card Categories	Card Categories	Card Categories	Qtr 2	Qtr 3	Qtr 4	Usage Changes	Usage Changes	Expense Items	Expense Items	Expense Items	Expense Items
708082083	20	0	87.0	01-2023	Week-1	3544.0	1661	86.875	111	0.469	1917.48750000	0	Fals	Fals	Fals	Fals	Fals	Fals	Fals	Fals	Fals	Fals	Fals	True
708083283	445	1	108.0	01-2023	Week-1	3421.0	2517	992.0	216	0.736	69.4	0	Fals	Fals	Fals	Fals	Fals	Fals	Fals	True	True	Fals	Fals	Fals
708084558	140	0	106.0	01-2023	Week-1	8258.0	1771	1447.0	234	0.214	202.58	0	Fals	Fals	Fals	Fals	Fals	Fals	Fals	Fals	Fals	Fals	Fals	Fals
708085458	250	1	150.0	01-2023	Week-1	1438.3	0	3940.0	820	0.64	236.0	0	Fals	Fals	Fals	Fals	Fals	Fals	True	Fals	Fals	Fals	Fals	Fals

Client - Num	Ann - Feb	Act - Mar	Cost - Apr	Week - May	Week - Jun	Credit - Jul	Total - Aug	Total - Sep	Total - Oct	Avg - Nov	Interest - Dec	Delinquency - Jan	Card - Feb	Card - Mar	Card - Apr	Qtr - May	Qtr - Jun	Qtr - Jul	Use - Aug	Use - Sep	Exp - Oct	Exp - Nov	Exp - Dec	Exp - Jan
708086958	320	1	1060	01-2023	Week-1	3128.0	749	69.0	59	0.239	10.87	0	False	False	False	False	False	False	False	True	False	False	True	False
708095133	100	0	94.0	01-2023	Week-1	2386.5	1833	1448.0	29	0.055	27.12	0	False	False	False	False	False	False	True	False	False	False	False	False
708098133	225	1	75.0	01-2023	Week-1	2834.0	1418	1598.0	39	0.95	15.8	0	False	False	False	False	False	False	True	False	False	False	False	False
708099183	400	1	75.0	01-2023	Week-1	5723.0	1873	2732.0	63	0.327	40.98	0	False	False	False	False	False	False	True	False	False	False	True	False
708100533	200	1	64.0	01-2023	Week-1	2679.0	2277	4943.0	85	0.85	98.6	0	False	False	False	False	False	False	False	False	False	True	False	False
708103608	95	1	80.0	01-2023	Week-1	1189.0	2517	8618.875	128	0.212	19.002	0	False	True	False	False	False	False	False	False	False	False	True	False
708104658	455	1	118.0	01-2023	Week-1	1438.3	890	2928.0	48	0.619	73.2	0	False	False	False	False	False	False	True	False	False	True	False	False

Client - Num	Ann - Fees	Activatio - Days	Customer - Acq - Cost	Week - Start - Date	Week - End - Date	Credit - Limit	Total - Revenue - Bill	Total - Transactions - Amount	Total - Transactions - Volume	Avg - Utiliza - tion - Ratio	Interest - Earned	Delin - quent - Amount	Card - Category - Gold	Card - Category - Platinum	Card - Category - Silver	Qtr - Q2	Qtr - Q3	Qtr - Q4	User - Churn - Rate	User - Churn - Rate - Swipe	Exp - Type - Entertainment	Exp - Type - Food	Exp - Type - Grocery	Exp - Type - Travel	
70810833	485	0	86.0	01-20-23	Week-1	5590.0	0	1507.0	320	0.7	0	0	False	False	False	False	False	False	False	True	True	False	False	False	False
708112008	440	1	86.0	01-20-23	Week-1	23510.0	1049	1661.0	355	0.045	465.08	0	False	False	False	False	False	False	True	False	False	False	True	False	
708113208	300	1	149.0	01-20-23	Week-1	1688.0	0	4375.0	690	0.5	520.0	0	False	False	False	False	False	False	False	False	False	True	False	False	
708117933	360	0	70.0	01-20-23	Week-1	1880.0	0	2469.0	340	0.9	4173.0	0	False	False	False	False	False	False	False	False	False	False	False	False	
708119658	340	0	44.0	01-20-23	Week-1	12836.0	1034	2519.0	531	0.081	554.18	0	False	False	False	False	False	False	True	False	False	False	False	False	
708121908	405	0	71.0	01-20-23	Week-1	22917.0	0	2045.0	450	0.5	265.85	0	False	False	False	False	False	False	True	False	False	False	True	False	
708123033	355	0	78.0	01-20-23	Week-1	11463.0	0	8618.8	1075	0.00	1917.48750000	0	False	False	True	False	False	False	True	False	False	True	False	False	

Task: regression | Best Model: Linear Regression

****Linear Regression****: rmse=1363749.4313, mae=559169.4088, r2=0.9986 ****Random Forest****: rmse=1442863.0916, mae=572068.4692, r2=0.9984 ****Gradient Boosting****: rmse=2123318.3993, mae=1569962.64, r2=0.9966

Deep Learning Results

Model: MLP | Task: regression

• rmse: 739737901.1898 • r2: -408.7242

Business Insights & Recommendations

LLM insight generation failed: Error code: 400 - {'error': {'message': 'The model `llama3-70b-8192` has been decommissioned and is no longer supported. Please refer to <https://console.groq.com/docs/deprecations> for a recommendation on which model to use instead.', 'type': 'invalid_request_error', 'code': 'model_decommissioned'}}